# WILSON

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**Hands-On Engineering Lead — Banking Transaction Platforms | Payments, Lending & Customer Data · Platform Modernisation**

## PROFESSIONAL SUMMARY

Engineering lead with 18+ years of software engineering experience, who leads teams while staying hands-on in architecture, solution design and code reviews. Since 2022, a Vice President in UOB's Group Technology & Operations, accountable for mission-critical payment and lending transaction platforms in a Tier-1, MAS-regulated bank.

Leads 25 onshore and offshore engineers across payment, customer-data and retail-lending platforms serving millions of accounts. Delivered the ISO 20022 (SWIFT MT-to-MX) migration and put AI-assisted code review into production.

Before UOB, built and re-architected consumer internet platforms as an engineer and technical lead at Renren, Sohu and Baidu, then held head-of-technology and architect roles at start-ups and scale-ups (2016–2022), including leading VV Technology's 140-person technology organisation through its managers. At UOB, reports status, risks and decisions to senior management, the steering committee and the business team.

## CORE COMPETENCIES

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| **Banking Domains** | Payment initiation, validation and status tracking (ISO 20022, SWIFT, RTGS), retail lending servicing and collections, customer-360 data, integration with core banking |
| **Engineering Leadership** | Leading squads through squad leads and engineering managers, coaching and mentoring, architecture and code reviews, contributed to bank-wide SDLC and code-review standards |
| **Architecture & Integration** | Spring Cloud microservices and API design, event-driven integration (Kafka), distributed high-concurrency systems, Apache Camel integration middleware, legacy modernisation and migration |
| **Hands-On Engineering** | Java (concurrency control, JVM tuning), Spring Boot, Apache Camel, Spring Batch, React, SQL, MySQL, Redis |
| **Cloud-Native & DevSecOps** | Kubernetes, Docker, AWS, Linux (RHEL); CI/CD on Jenkins and Bitbucket with a 99% SonarQube coverage gate on all code, automated regression and penetration testing; secure SDLC |
| **Reliability & Performance** | Disaster recovery (DR) and failover drills, high availability, latency tuning under high load, Splunk monitoring, P1–P3 incident response and root-cause analysis (RCA) |
| **AI-Assisted Development** | Production AI code review on headless Claude Code agents, human-in-the-loop governance, prompt A/B evaluation |

## SELECTED ACHIEVEMENTS

- **Core-banking integration:** Lead the teams behind two platforms integrated with core banking: a payment-initiation hub that hands validated payments to core banking, which executes settlement, and processes the results returned; and a regional customer-360 platform that takes daily account summaries from core banking for portfolio analysis.
- **Hands-on core code:** Personally built the hub's core modules — payment-transaction initiation, concurrency control, the corporate- and user-role permission matrix and processing of payment status results returned by downstream systems.
- **Legacy modernisation:** Planned and architected the phased move of cross-border payments to ISO 20022 (SWIFT's MT-to-MX migration), running old and new message formats side by side: new-format messaging went live in May and August 2025, ahead of SWIFT's November 2025 deadline, and the old path was retired in August 2026 with no service interruption.
- **Lending servicing and collections:** Own technology for retail unsecured-lending servicing — payment initiation, validation and lifecycle tracking — and post-disbursement collections across credit-card and unsecured-loan portfolios.
- **AI-assisted development:** Designed and built a production AI code-review service, live since May 2026 after formal architecture and risk approval: headless Claude Code agents review each pull request alongside SonarQube, cutting senior engineers' review time by an estimated ~80%; the AI never approves or merges.
- **Payment resilience:** Defined the DR strategy, failover procedures and capacity baselines for real-time payment middleware and personally led DR and failover drills, sustaining 99.9%+ availability under peak load.

## PROFESSIONAL EXPERIENCE

### United Overseas Bank (UOB) | Singapore — Oct 2022 – Present

*Vice President (VP2) / Technical Delivery Manager — Real-Time & Payment Data Services, Group Technology & Operations*

Technology leader for two mission-critical platform lines in a MAS-regulated Tier-1 bank — cross-border payments (SWIFT ISO 20022 messaging) and retail unsecured lending — serving millions of accounts.

***Architecture & Engineering***

- Led end-to-end design of a payment-initiation hub on Java, Spring Boot and Apache Camel — API contracts, orchestration, validation, enrichment and status tracking — integrating digital and branch channels, reference data, and vendor and third-party banking systems.
- Stay hands-on alongside delivery: review newly submitted core pull requests daily, co-design solutions with engineers, own the architecture, write core modules for significant changes and resolve production issues.
- Reduced end-to-end transaction latency under high load through thread-pool and JVM tuning and removal of synchronous blocking.
- Lead the 10-engineer backend team of a regional customer-360 platform, in production since 2022: evaluated Cloudera as the processing framework, designed the data models and personally built the data-analysis module and Spring Batch jobs.

***Delivery & Platform Ownership***

- Led a 15-engineer onshore/offshore team and owned end-to-end delivery of the ISO 20022 programme — scope, roadmap, risks and dependencies, release planning and production rollout — shipping upgrades every 2–3 months after go-live.
- Resolve schedule conflicts between the two platform lines by agreeing MVP scope with the requirement, project and technical leads, confirming dependencies and setting a sequence all parties accept.
- Act as the primary interface between business stakeholders, product owners, operations, compliance, architects and engineering managers.
- Hold decision or joint-decision authority on technology investment, vendor selection and SOW/licence negotiation, and chair governance of four IT service vendors — periodic budget reviews, year-end performance reviews and accountability meetings.

***Reliability, Security & Governance***

- Coordinate P1–P3 incidents end to end — escalation, stakeholder communication, RCA and post-incident reports; with no separate SRE team, the platform teams own production support, service-level indicators/objectives and Splunk monitoring.
- Led API encryption, database encryption of sensitive data, credential management and data migration for the payment, lending and customer-data platforms under MAS Technology Risk Management (TRM) guidelines.
- Personally prepare the change request for every production release on both platform lines and present the go-live plan, risk controls, rollback plan, test results and database changes to the Change Advisory Board (CAB).
- Established delivery KPIs — deployment frequency, mean time to recovery, incident trend and change-failure rate — and contributed to bank-wide SDLC, coding and code-review standards and to architecture-review governance.

### Golden Bridge Drive (SG) Pte. Ltd. | Singapore — Nov 2020 – Jul 2022

*Project Manager & Architect*

Multi-tenant SaaS for online-game businesses, built 0-to-1 on AWS — KYC, bonus incentives, VIP tiers and multi-channel payments.

- Architected and delivered the platform on Spring Cloud and Kubernetes behind an API gateway, including hierarchical KYC identity verification and rule models for authentication levels and bonus eligibility, triggers and distribution.
- Led a 40-person project team, including outsourced staff, introduced Agile/Scrum from scratch and CI on Jenkins and Amazon ECR, and owned delivery budget, capacity and launch readiness.

### VV Technology Pte. Ltd. | Singapore — Apr 2019 – Jun 2020

*Head of Technical Centre*

VV-Life: a local-life services platform connecting offline merchants — food, fresh produce, travel and everyday services.

- Led VV's 140-person technology organisation through its managers and built its microservices platform from scratch, owning technology selection and the technology budget.
- Delivered the first VV-Life release in 2019, including the payment subsystem — transaction, management and settlement-reconciliation modules — which integrated UnionPay, Visa/Mastercard, Google Pay and GrabPay payments.
- Designed for performance with Spring Cloud microservices, Kafka-based asynchronous processing and caching that kept load off the database, bringing service response times to the millisecond level.
- Introduced CI/CD, code standards and containerised deployment with rollback on Kubernetes; defined production-incident priority levels and tiered emergency-response plans.

### AI Park | China — Apr 2016 – Sep 2017

*VP of Technical Department*

- Led architecture, delivery, budget and the technical team for a smart-parking IoT platform connecting car parks to the cloud for automated management, online payment and recommendations.

- Brought Spring Cloud microservices into the architecture, with Kafka for asynchronous flows and caching to shield the database, achieving millisecond-level service processing, and led the big-data team.

## EARLIER EXPERIENCE

- **Hengshunyuan Information Technology — Co-founder (Oct 2017 – Sep 2019, China):** Owned product, architecture and delivery of an audio-learning platform, reaching a peak of ~100k daily active users.
- **Baidu — Technical Specialist, Nuomi group-buying e-commerce platform (May 2014 – Dec 2014, China):** As project leader, re-architected the merchant-side order system on Dubbo microservices and delivered two months ahead of plan; designed and coded Nuomi's payment-gateway integration with Alipay, WeChat Pay and UnionPay.
- **Sohu.com — Technical Leader / Project Leader, Sohu News content platform (Mar 2012 – May 2014, China):** Led the re-architecture of the fifth-generation CMS onto Spring MVC and a Redis cluster, and delivered its OpenAPI messaging platform.
- **Renren Network — Senior R&D Engineer (Jul 2008 – Mar 2012, China):** Developed Renren's shopping website, Renren Mall and CRM, and designed and coded Renren Mall's payment integration with Alipay and UnionPay.

## EDUCATION, CERTIFICATIONS & LANGUAGES

- Master's degree in Computer Science and Technology — Liaoning University (2005 – 2008)
- Bachelor's degree in Applied Mathematics — Shenyang University of Technology (2000 – 2004)
- Software Designer (Intermediate) — Qualification Certificate of Computer and Software Technology Proficiency (2006) · PMP — in progress · Site Reliability Engineering (SRE) — formal training
- Languages: English and Mandarin (professional working proficiency in both)